Temozolomide with Radiotherapy (RT) and Adjuvant Therapy

INDICATIONS FOR USE:

INDICATION	ICD10	Regimen Code	HSE approved reimbursement status*
Adult patients with newly-diagnosed glioblastoma multiforme concomitantly with radiotherapy (RT) followed by adjuvant treatment commencing 4 weeks after completion of chemoradiation	C71	00334a	N/A

* This applies to post 2012 indications

TREATMENT:

The starting dose of the drugs detailed below may be adjusted downward by the prescribing clinician, using their independent medical judgement, to consider each patients individual clinical circumstances.

Temozolomide is administered orally once daily with concomitant radiotherapy for 6 weeks.

Four weeks after completion of the temozolomide and radiotherapy concomitant phase:
Temozolomide is administered orally once daily on days 1-5 of a 28 day cycle for up to 6 cycles of monotherapy treatment or until disease progression or unacceptable toxicity develops.

Temozolomide and Concomitant Radiotherapy (RT) Phase			
Drug	**Dose**	**Route**	**Cycle**
Temozolomide	75mg/m^2 ONCE daily concomitantly with radiotherapy	PO[a]	Continuously with radiotherapy for 6 weeks
PJP prophylaxis **(Refer to local policy)** is required when temozolomide is administered with concomitant radiotherapy			
4 WEEK BREAK FOLLOWED BY:			
Temozolomide Monotherapy Phase (Adjuvant)			
Day	**Drug**	**Dose**	**Route**	**Cycle (28 days)**
1-5	Temozolomide	150mg/m^2 ONCE daily	PO	1
1-5	Temozolomide	200mg/m^2 ONCE daily [b]	PO	2-6
[a]Temozolomide is available as 5 mg, 20 mg, 100 mg, 140 mg, 180 mg, or 250 mg hard capsules				
Temozolomide hard capsules should be administered in the fasting state. The capsules must be swallowed whole with a glass of water and must not be opened or chewed. If vomiting occurs after the dose is administered, a second dose should not be administered that day				
If a dose is missed, the patient should make up that dose, unless the next dose is due within 12 hours.				
[b]See Dose Modifications: Temozolomide Monotherapy Phase below				

ELIGIBILITY:

- Indications as above
- ECOG 0-2
- Adequate renal and hepatic function

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EXCLUSIONS:

- Patients with hypersensitivity to temozolomide or any of its listed excipients
- Hypersensitivity to dacarbazine
- Severe myelosuppression
- Significant hepatic dysfunction
- Pregnancy or breastfeeding

PRESCRIPTIVE AUTHORITY:

The treatment plan must be initiated by a Consultant Medical Oncologist

TESTS:

Baseline tests:

- FBC, renal and liver profile
- Blood glucose
- Virology screen -Hepatitis B* (HBsAg, HBcoreAb)

*(Reference Regimen Specific Complications for information on Hepatitis B reactivation)

Regular tests:

Temozolomide and Concomitant Radiotherapy (RT)

- FBC profile weekly
- Renal and liver profile before week 4

Temozolomide Monotherapy Phase (Adjuvant)

- FBC, renal and liver profile every 28 days (at day 1 of each cycle)

Disease monitoring:

Disease monitoring should be in line with the patient's treatment plan and any other test/s as directed by the supervising Consultant.

DOSE MODIFICATIONS:

Temozolomide and Concomitant Radiotherapy (RT)

- Any dose modification should be discussed with a Consultant.
- No dose reductions are recommended, but delay or discontinuation of temozolomide administration should be decided weekly according to haematological and non-haematological toxicity criteria.
- Temozolomide administration can be continued throughout the 42 day concomitant period (up to 49 days) if all of the following conditions are met:
  - absolute neutrophil count (ANC) ≥ 1.5 x 10^9/L
  - platelet count ≥ 100 x 10^9/L
  - common toxicity criteria (CTC) non-haematological toxicity ≤ Grade 1 (except for alopecia, nausea and vomiting).
- During treatment with concomitant RT a complete blood count should be obtained weekly.
- Temozolomide administration should be temporarily interrupted or permanently discontinued during the concomitant phase according to the haematological and non-haematological toxicity criteria as noted in Table 1**.**

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Table 1: Temozolomide dosing interruption or discontinuation during concomitant radiotherapy and temozolomide

Toxicity	Temozolomide Interruption	Temozolomide Discontinuation
ANC	≥ 0.5 and <1.5 x 10^9/L	< 0.5 x 10^9/L
Platelets	≥ 10 and <100 x 10^9/L	<10 x 10^9/L
CTC non-haematological toxicity (except for alopecia, nausea, vomiting)	CTC Grade 2	CTC Grade 3 or 4

Temozolomide Monotherapy Phase

- Any dose modification should be discussed with a Consultant
- Dose in Cycle 1 is 150 mg/m^2 once daily for 5 days followed by 23 days without treatment.
- At the start of Cycle 2, the dose is escalated to 200 mg/m^2 if the CTC non-haematological toxicity for Cycle 1 is Grade ≤ 2 (except for alopecia, nausea and vomiting), absolute neutrophil count (ANC) is ≥ 1.5 x 10^9/L, and the platelet count is ≥ 100 x 10^9/L.
- If the dose was not escalated at Cycle 2, escalation should not be done in subsequent cycles.
- Once escalated, the dose remains at 200 mg/m^2 per day for the first 5 days of each subsequent cycle except if toxicity occurs.

Table 2: Temozolomide dose levels for monotherapy treatment

Dose Level	Temozolomide Dose (mg/m^2)	Remarks
-1	100	Reduction for prior toxicity
0	150	Dose during Cycle 1
1	200	Dose during Cycles 2-6 in absence of toxicity

Table 3: Temozolomide dose reduction or discontinuation during monotherapy treatment

Toxicity	Reduce temozolomide by 1 dose level[a]	Discontinue Temozolomide
ANC	< 1 x 10^9/L	See footnote b
Platelets	< 50 x 10^9/L	See footnote b
CTC non-haematological toxicity (except for alopecia, nausea, vomiting)	CTC Grade 3	CTC Grade 4[b]

[a]Temozolomide dose levels are listed in Table 2.

[b]Temozolomide is to be discontinued if:

- dose level -1 (100 mg/m^2) still results in unacceptable toxicity
- the same Grade 3 non-haematological toxicity (except for alopecia, nausea, vomiting) recurs after dose reduction.

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Renal and Hepatic Impairment:

Table 4: Dose modification of temozolomide in renal and hepatic impairment

Renal Impairment		Hepatic Impairment	
CrCl (mL/minute)	**Dose**	**Impairment level:**	**Dose**
≥ 36	No dose adjustment is needed	Child-Pugh A/B:	No dose adjustment is needed
< 36	No need for dose adjustment is expected	Child-Pugh C:	No need for dose adjustment is expected
Haemodialysis	No need for dose adjustment expected		
Recommendations as per Giraud et al 2023			

SUPPORTIVE CARE:

EMETOGENIC POTENTIAL:

- As outlined in NCCP Classification Document for Systemic AntiCancer Therapy (SACT) Induced Nausea and Vomiting Available on the NCCP website

Temozolomide: Moderate to high (**Refer to local policy).**

For information:

Within NCIS regimens, antiemetics have been standardised by the Medical Oncologists and Haemato-oncologists and information is available in the following documents:

- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Medical Oncology) - Available on the NCCP website
- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Haemato-oncology) - Available on the NCCP website

PREMEDICATIONS: None

OTHER SUPPORTIVE CARE:

PJP prophylaxis during concomitant RT phase **(Refer to local policy)**

Temozolomide can have genotoxic effects. Therefore, men being treated with it should be advised not to father a child up to 6 months after receiving the last dose and to seek advice on cryoconservation of sperm prior to treatment, because of the possibility of irreversible infertility due to therapy with temozolomide.

ADVERSE EFFECTS

- Please refer to the relevant Summary of Product Characteristics (SmPC) for details.

REGIMEN SPECIFIC COMPLICATIONS

- **Hepatitis B Virus (HBV):** Hepatitis due to HBV reactivation, in some cases resulting in death, has been reported. Experts in liver disease should be consulted before treatment is initiated in patients with positive hepatitis B serology (including those with active disease). During treatment patients should be monitored and managed appropriately.

DRUG INTERACTIONS:

- Current SmPC and drug interaction databases should be consulted for more information.

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1. Stupp R, Mason WP, van den Bent MJ, et al. Radiotherapy plus concomitant and adjuvant temozolomide for glioblastoma. N Engl J Med 2005;352(10):987-96.
2. Stupp R, Hegi M. & Mason W. et al. Effects of radiotherapy with concomitant and adjuvant temozolomide versus radiotherapy alone on survival in glioblastoma in a randomised phase III study: 5-year analysis of the EORTC-NCIC trial. Lancet Oncol. 2009:10 (5):459-66.
3. Giraud E L, Lijster B D, et al. Dose recommendations for anticancer drugs in patients with renal or hepatic impairment: an update. Available at: https://pubmed.ncbi.nlm.nih.gov/37269847/
4. NCCP Classification Document for Systemic Anti-Cancer Therapy (SACT) Induced Nausea and Vomiting. V6 2025. Available at: https://www.hse.ie/eng/services/list/5/cancer/profinfo/chemoprotocols/nccp-classification-document-for-systemic-anti-cancer-therapy-sact-induced-nausea-and-vomiting.pdf
5. Temozolomide (Temodal ®) Summary of Product Characteristics Accessed July 2025. Available at: https://www.ema.europa.eu/en/documents/product-information/temodal-epar-product-information_en.pdf

Version	Date	Amendment	Approved By
1	20/06/2016		Prof Maccon Keane
2	20/06/2018	Updated with new NCCP regimen template and clarified treatment table	Prof Maccon Keane
3	15/07/2020	Regimen review Updated emetogenic potential	Prof Maccon Keane
4	23/07/2025	Reviewed. Updated exclusions section. Renal and Hepatic dose modifications updated to align with Giraud et al 2023. Regimen updated in line with NCCP standardisation.	Prof Maccon Keane

Comments and feedback welcome at oncologydrugs@cancercontrol.ie.

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